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Evaluation of global leaf area index simulated by LM4+, the land component of KIOST-ESM2

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INTRODUCTION

Background: KIOST is developing KIOST-ESM2 for CMIP7 with GFDL's LM4+ (LM4.1 family) as its land component. LM4+ grows vegetation as size-structured cohorts (PPA), so leaf area, biomass and canopy structure evolve with climate.
Vegetation controls the land–atmosphere exchange of energy, water and carbon; a biased leaf area index (LAI) therefore feeds back on the coupled climate. Offline evaluation under observed forcing isolates the land model's own errors.
Key question: Does offline LM4+ reproduce the magnitude and the seasonal timing of global LAI, and what causes the mismatch?
Objective 1: Benchmark 1982–2010 LAI against GIMMS LAI4g (pattern, magnitude, seasonal cycle).
Objective 2: Trace the autumn leaf-off delay to the phenology rule and test a day-length cue in pilot runs.

DATA & METHODS

Dataset	Setup / variables	Period / resolution
LM4+ offline (ESM4.5 lm4P)	PPA cohorts, 6 species, fixed species map, N cycle off, CO_2 fixed	1979–2023, C96 (≈1°) → 1°
WFDE5 forcing (Cucchi et al. 2020)	ERA5 bias-corrected with CRU/GPCC; T, q, wind, SW, LW, precip.	1979–2023, 0.5° hourly → C96
Initial land state	KIOST-ESM2 ≈400-yr coupled piControl + 1 cycle WFDE5 1981–2010	start 1979-01-01
GIMMS LAI4g (Cao et al. 2023)	AVHRR LAI, consolidated with MODIS	1982–2010, 15-day 1/12° → monthly 1°

Evaluation: 1982–2010 monthly climatology on a common 1° grid; MJJAS mean and bias; seasonal cycles in six IPCC AR6 regions. SOS/EOS = dates at which the interpolated monthly cycle crosses 50 % of its amplitude.
Pilot runs (year 2010 from the 1 Jan 2010 CTRL state): TC10 raises the leaf-off temperature threshold 5 → 10 °C; DL14+TC10 adds a day-length (< 14 h) leaf-off condition (code patch; larch and C3 grass).

RESULTS 1: GROWING-SEASON LAI

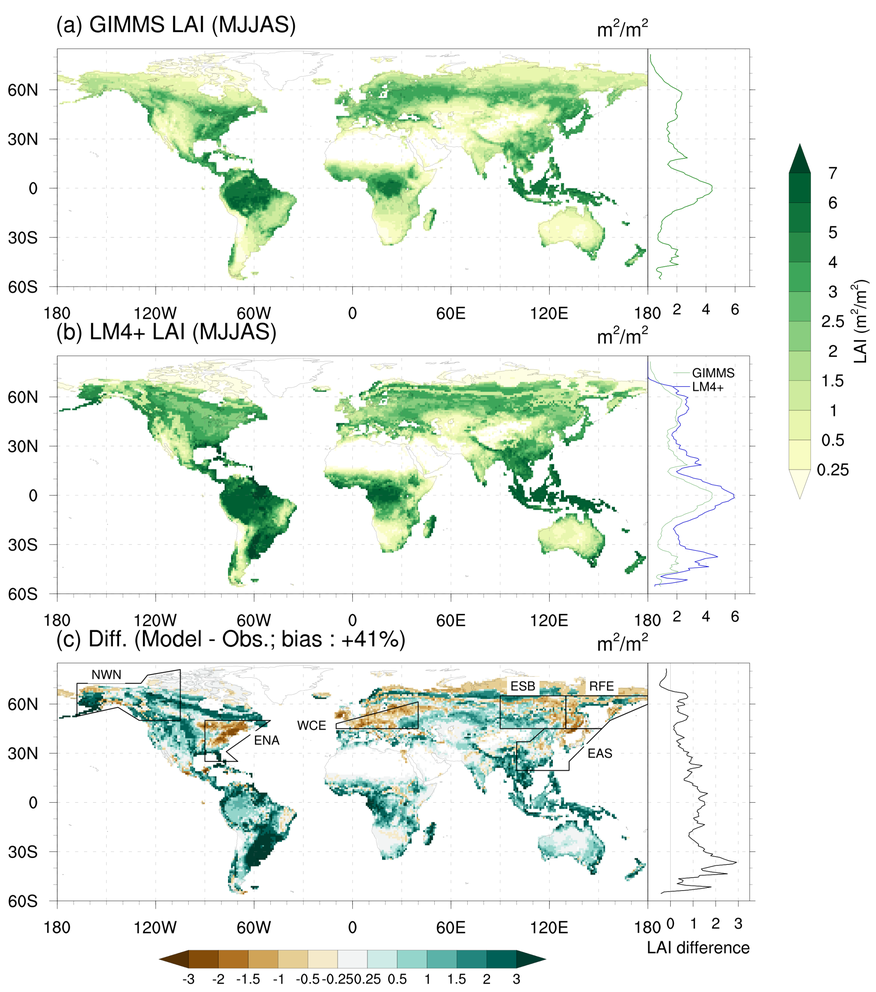


Figure 1. Growing-season (MJJAS) LAI, 1982–2010: (a) GIMMS LAI4g, (b) LM4+, (c) LM4+ − GIMMS. Side panels: zonal means. Polygons: the AR6 regions used in Figs. 2–3.

- The large-scale pattern is reproduced, but LAI is too high almost everywhere: global MJJAS mean 2.77 vs 1.97 (+41 %); annual +54 %.
- Largest excess over tropical and subtropical South America and Africa.
- **Boreal forest (50–70°N): the summer peak is only +15 % too high, yet the annual mean is +56 % too high — 87 % of the bias comes from the off-season (Sep–Apr). The extratropical excess is a timing problem.**

RESULTS 2: SEASONAL TIMING AND ITS CAUSE

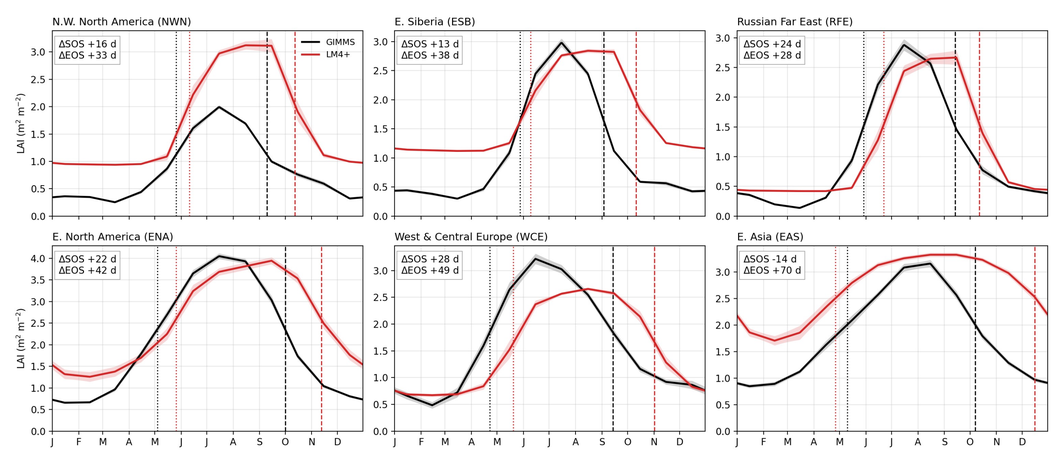


Figure 2. Climatological LAI seasonal cycle, 1982–2010, GIMMS vs LM4+ in six AR6 regions; markers: SOS/EOS at 50 % of the amplitude.

- **Leaf-off (EOS) is 28–70 days too late in every region (NWN +33, ESB +38, RFE +28, ENA +42, WCE +49, EAS +70 d); boreal leaf-on is also late (+13–24 d).**
- Cause: leaves are dropped only when the 20-day smoothed canopy temperature falls below 5 °C (10 °C for broadleaf trees). Daily T < 5 °C is reached ≈20 d after the observed EOS and the smoothing adds ≈17 d: ESB observed 3 Sep → model 11 Oct.
- The observed EOS instead clusters at a day length of 11.4–13.3 h regardless of temperature: a day-length cue is missing.

RESULTS 3: DAY-LENGTH LEAF-OFF PILOT RUNS

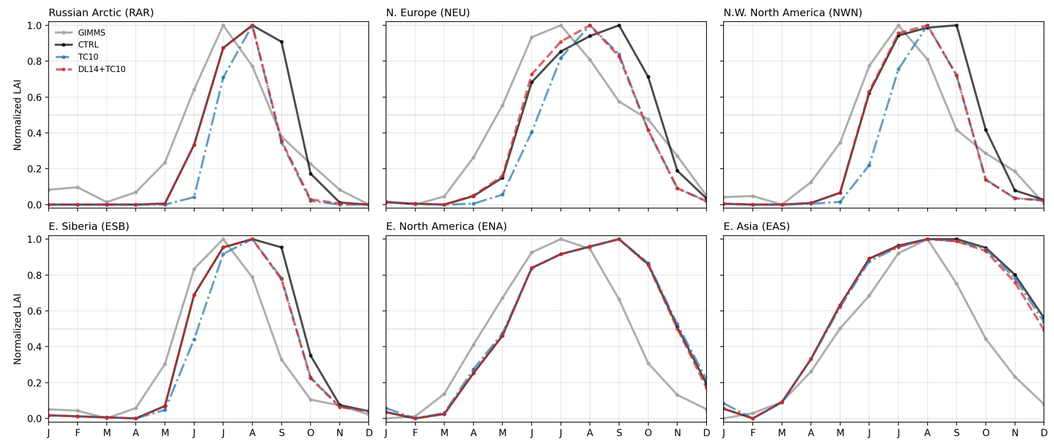


Figure 3. Normalized 2010 LAI cycle, (LAI − min)/(max − min), in six regions: GIMMS, CTRL, TC10 and DL14+TC10.

Temperature threshold alone (TC10): the autumn drop moves little and spring leaf-on is delayed as a side effect.
DL14+TC10 removes 20–50 % of the Sep–Oct excess in boreal regions (NWN 38 %, ESB 20 %, RFE 50 %); the 50–70°N EOS bias falls from +33 to +16 d with spring unchanged.
Temperate regions are unchanged: the rule was applied only to larch and C3 grass; broadleaf trees need a ≈11.5 h cue.
Even the small TC10 shift changes Sep–Oct fluxes at 50–70°N: ET −4.5 %, GPP −11 %, runoff +4.7 %.

SUMMARY & CONCLUSIONS

1. Offline LM4+ reproduces the global LAI pattern but overestimates it (+41 % in MJJAS); in the extratropics the excess comes mainly from a 1–2 month delay of autumn leaf-off.
2. The delay follows from a temperature-only phenology rule (5 °C threshold on a 20-day smoothed temperature); observations point to day length as the missing cue.
3. A day-length leaf-off rule removes about half of the boreal EOS bias with no spring side effect; type-specific thresholds (larch 13.5 h, C3 grass 12.5 h, broadleaf 11.5 h) are being tested.
4. Next: replant species from an observed map (larch is nearly absent in the current state), time-varying CO_2 and the C–N version, then coupled KIOST-ESM2 experiments.

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